

Expense Tracker - Complete Technical Documentation

Date: March 31, 2026 Project Root: `expense-tracker`

1. Executive Summary

This application is a full-stack Expense Tracker designed for corporate trip and reimbursement workflows.

It supports:

- authentication and role-based access (submitter, approver, admin)
- trip planning with itinerary segments
- expense submission with receipt storage
- advance request and approval cycles
- admin approval/rejection workflows
- analytics dashboards from live transactional data

The backend is Node.js + Express + PostgreSQL, and the frontend is Vue 3 (Vite, Pinia, Vue Router).

2. Goals and Scope

This document covers:

- complete working of the application
- architecture and code-level module explanation
- database schema with ER diagram
- route, controller, model, and frontend flow exploration
- screenshots of key application screens
- setup and runtime process

This document does not include:

- CI/CD pipeline implementation
- production infrastructure provisioning
- cloud deployment scripts

3. Technology Stack

Backend

- Node.js
- Express
- PostgreSQL (`pg`)
- Session storage via `connect-pg-simple`
- Authentication with `express-session` + `bcrypt`
- File upload with `multer` (memory storage) and persisted as BYTEA

Frontend

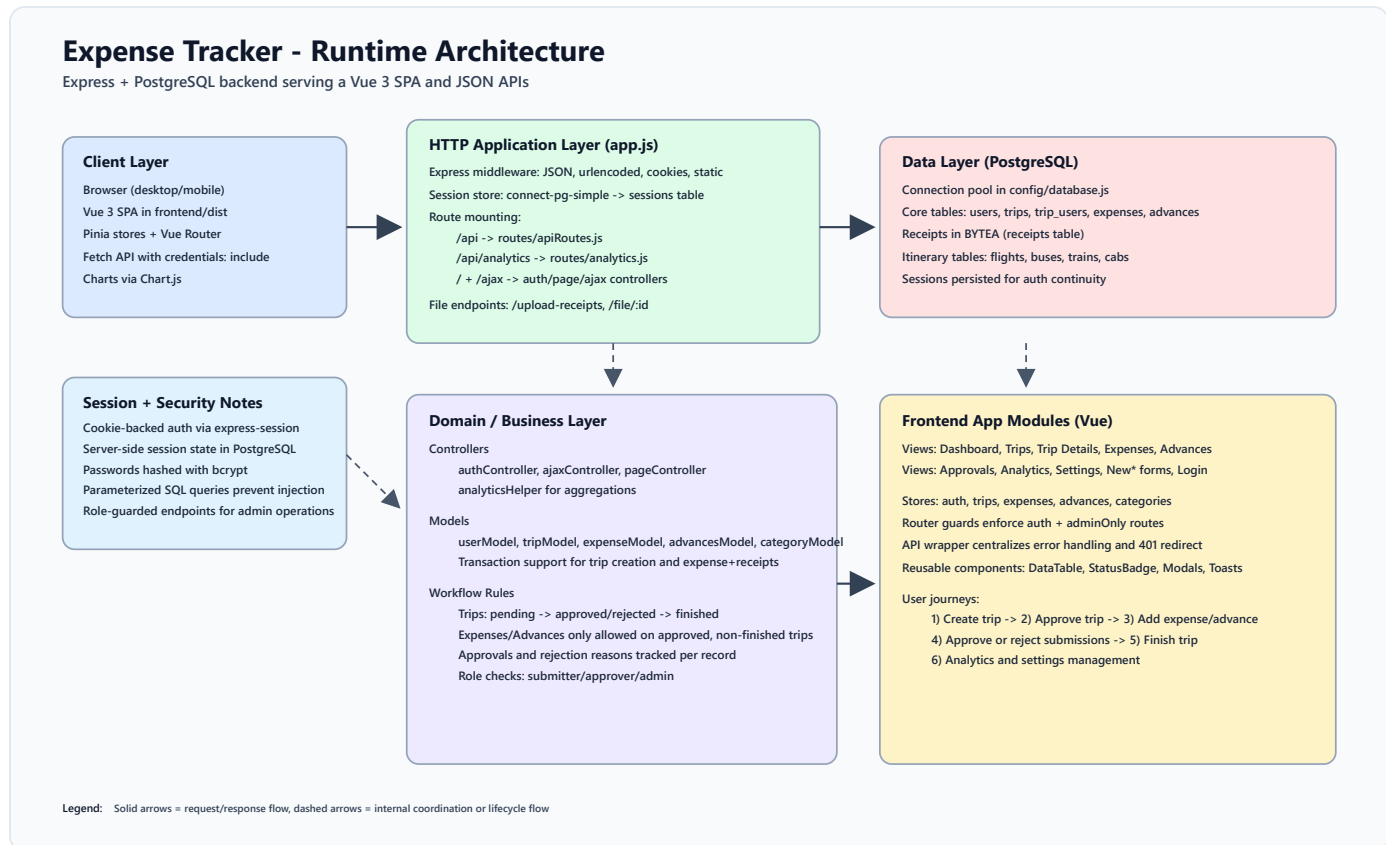
- Vue 3
- Vue Router
- Pinia
- Chart.js
- Bootstrap Icons

- Vite

Database

- PostgreSQL with relational schema
- transactional writes for complex operations
- foreign keys, constraints, and indexes

4. High-Level Architecture



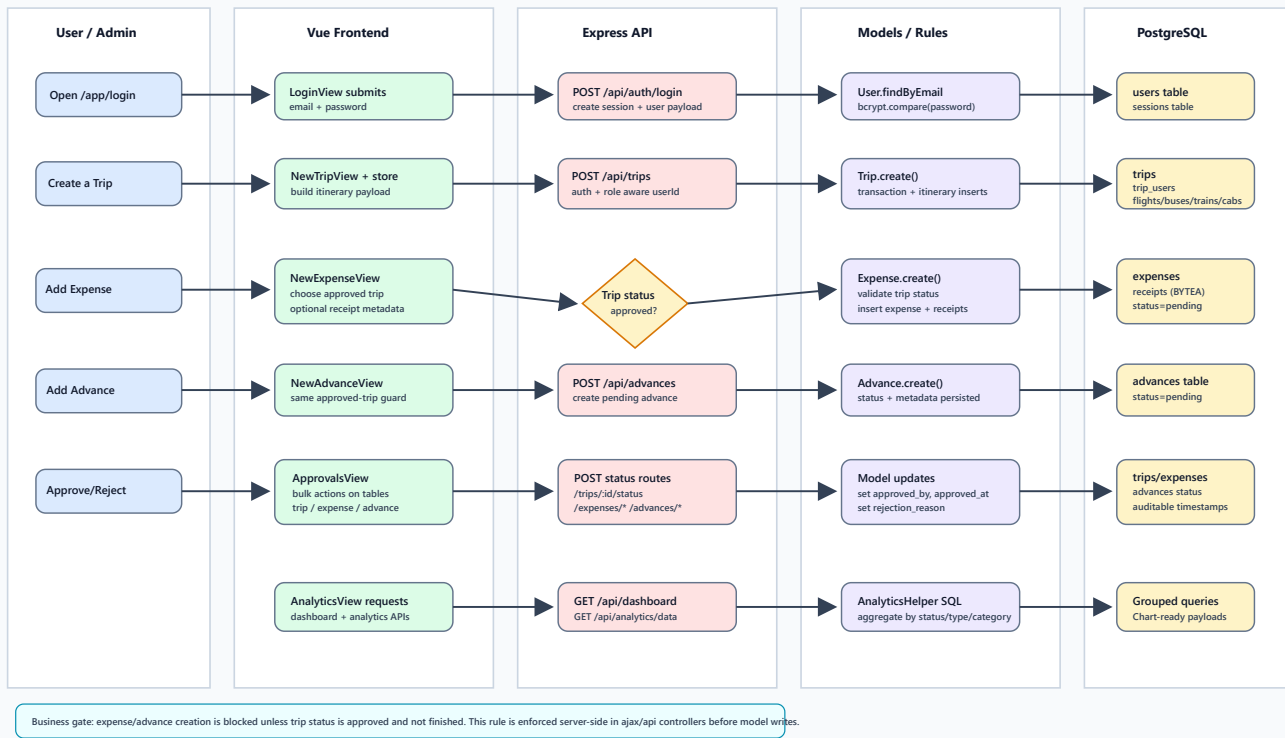
Architecture Notes

1. Browser loads Vue SPA from backend-served static assets (frontend/dist) if built.
2. Frontend calls JSON APIs under /api/* with session cookies.
3. Express handles auth/session checks and delegates to controllers.
4. Controllers use model classes for domain behavior and validation.
5. Models execute parameterized SQL using a pooled PostgreSQL connection.
6. Session state is persisted in PostgreSQL (sessions table).

5. Full Working Flow

End-to-End Working Flow

From login to trip approval, expense lifecycle, and analytics rendering



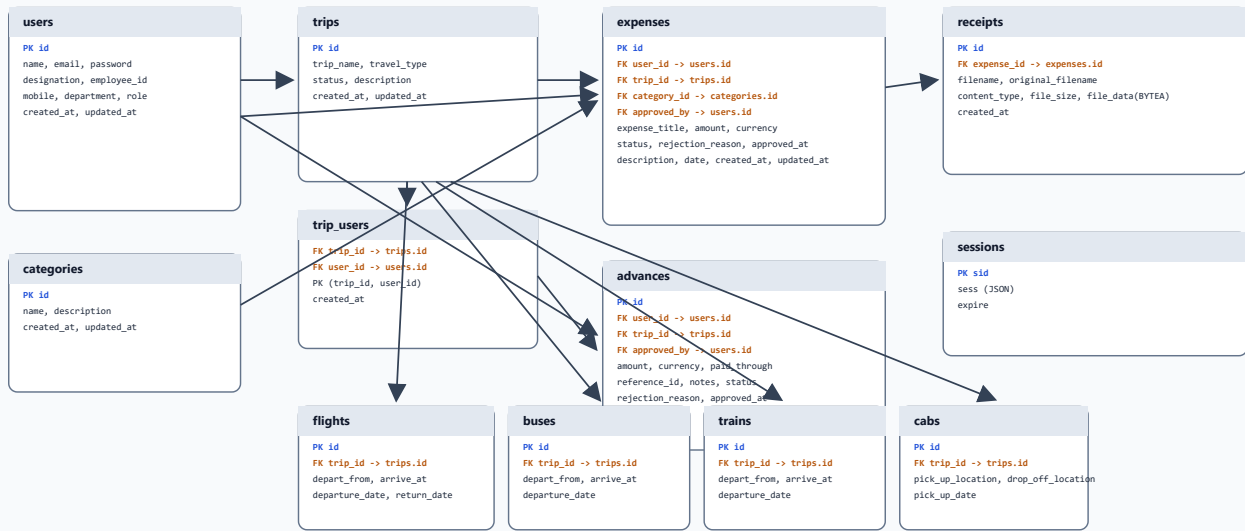
Core Lifecycle

1. User logs in via `/api/auth/login`.
2. Session is created and persisted.
3. Submitter creates trips (status starts as `pending`).
4. Admin approves/rejects trips.
5. Expenses and advances can only be created for `approved` trips.
6. Admin reviews and approves/rejects expense and advance items.
7. Trip can be moved to `finished` after processing.
8. Analytics endpoints aggregate and return chart-ready data.

6. Database Design and ER Diagram

Expense Tracker - Entity Relationship Diagram

Current PostgreSQL model used by app.js, models/*, and API routes



Legend and Notes

PK = Primary Key, FK = Foreign Key, all FK constraints use referential integrity in PostgreSQL.

Trip membership is many-to-many via trip_users. Receipts are 1-to-many from expenses and stored as BYTEA.

Approval workflow fields (status, approved_by, approved_at, rejection_reason) are present on expenses and advances.

Sessions table is used by connect-pg-simple for Express session persistence and is not directly joined with domain tables.

Main Tables

- **users** : identity, department, designation, role
- **categories** : expense categories
- **trips** : trip request metadata and status
- **trip_users** : many-to-many mapping between trips and users
- **flights** , **buses** , **trains** , **cabs** : normalized itinerary segments
- **expenses** : expense transactions with approval fields
- **receipts** : binary receipt files linked to expenses
- **advances** : advance transactions with approval fields
- **sessions** : express-session persistence table

Important Relationships

- `trip_users.trip_id -> trips.id`
- `trip_users.user_id -> users.id`
- `expenses.user_id -> users.id`
- `expenses.trip_id -> trips.id`
- `expenses.category_id -> categories.id`
- `receipts.expense_id -> expenses.id`
- `advances.user_id -> users.id`
- `advances.trip_id -> trips.id`

7. Backend Code Exploration

7.1 App Bootstrap (`app.js`)

Responsibilities:

- load env variables
- create Express app
- configure middleware and session storage
- mount API and web routes
- expose upload and file endpoints
- serve Vue build and fallback for SPA routes

Notable behavior:

- checks PostgreSQL connectivity at startup
- mounts `/api`, `/api/analytics`, `/`, `/ajax`
- uses `connect-pg-simple` with shared DB pool

7.2 Authentication and Session

Primary flow:

- Login request hits `POST /api/auth/login`
- User lookup through `User.findByEmail`
- Password check with `bcrypt.compare`
- Session object stores user profile and role
- Route guards validate `req.session.isAuthenticated`

Route guard behavior (`authMiddleware.js`):

- returns JSON `401` for API requests
- redirects browser requests to `/app/login`

7.3 API Layer (`routes/apiRoutes.js`)

Major endpoint groups:

- Auth: `/auth/login`, `/auth/logout`, `/auth/me`
- Users: CRUD with admin restrictions
- Trips: list, create, status update, finish
- Expenses: CRUD + approve/reject
- Advances: CRUD + approve/reject
- Categories, dashboard, analytics integration

7.4 Controllers

- `authController.js` : login/logout/index redirects for page routes
- `ajaxController.js` : legacy AJAX endpoints plus approval actions
- `pageController.js` : redirect-style page route compatibility
- `analyticsHelper.js` : SQL aggregations for charts and summaries

7.5 Model Layer

Each model wraps table operations and business-safe patterns:

- `userModel.js`
- `tripModel.js`
- `expenseModel.js`
- `advancesModel.js`
- `categoryModel.js`

Design choices:

- parameterized SQL for injection safety
- transaction helper for multi-table writes
- centralized connection/query helpers in `config/database.js`

7.6 Key Business Rules in Code

1. Trips default to `pending`.
2. Only admins can approve/reject/finish trips.
3. Expenses and advances are only allowed on approved trips.
4. Pending items can be approved or rejected with a reason.
5. Finished trips are terminal for new submissions.

8. Frontend Code Exploration

8.1 App Startup

- `frontend/src/main.js` boots Vue + Pinia + Router.
- `App.vue` performs initial `auth.checkAuth()`.

8.2 Router and Guards

`frontend/src/router/index.js` defines:

- guest route: `/app/login`
- authenticated shell: `/app/*`
- admin-only route: `/app/approvals`

Guard logic:

- load auth state before routing
- redirect unauthenticated users to login
- block admin-only routes for non-admin users

8.3 State Management (Pinia)

Stores:

- `auth.js` : login/logout/session user state
- `trips.js` : list/detail/status operations
- `expenses.js` : expense CRUD and approvals
- `advances.js` : advance CRUD and approvals
- `categories.js` : category management

8.4 API Client

`frontend/src/api/index.js` :

- central fetch wrapper
- `credentials: include` for session cookies
- uniform error handling
- auto-redirect on 401 (except auth checks)

8.5 Views and UX Modules

Primary views:

- Login, Dashboard
- Trips, New Trip, Trip Details
- Expenses, New Expense

- Advances, New Advance
- Approvals, Analytics, Settings

Shared components:

- layout shell (`AppLayout`)
- data table, status badges, modals
- toasts for feedback

9. Approval and Expense Workflow Details

Trip Workflow

- Create trip -> `pending`
- Admin action -> `approved` or `rejected`
- Once reconciled -> `finished`

Expense Workflow

- Expense creation validates trip state
- New expense status: `pending`
- Admin can approve in bulk or reject with reason

Advance Workflow

- Same approval lifecycle as expenses
- supports payment metadata (`paid_through` , `reference_id`)

10. Analytics Implementation

Analytics uses SQL aggregations in `analyticsHelper.js` :

- expense totals by category
- trip counts by travel type
- advance totals by currency
- user-scoped and global (admin) views

Frontend renders results using Chart.js in `AnalyticsView.vue` .

11. Security and Data Integrity

Security controls:

- bcrypt password hashing
- session-based authentication
- role checks for privileged actions
- parameterized SQL statements

Integrity controls:

- foreign keys and constraints
- status checks in route/model logic
- transaction wrapping for multi-table writes

12. Setup and Run Instructions

Prerequisites

- Node.js
- PostgreSQL running locally

Install and run

```
npm install
cd frontend && npm install
```

Database setup (recommended)

```
node database/setup-interactive.js
```

Start app

```
npm start
```

Open:

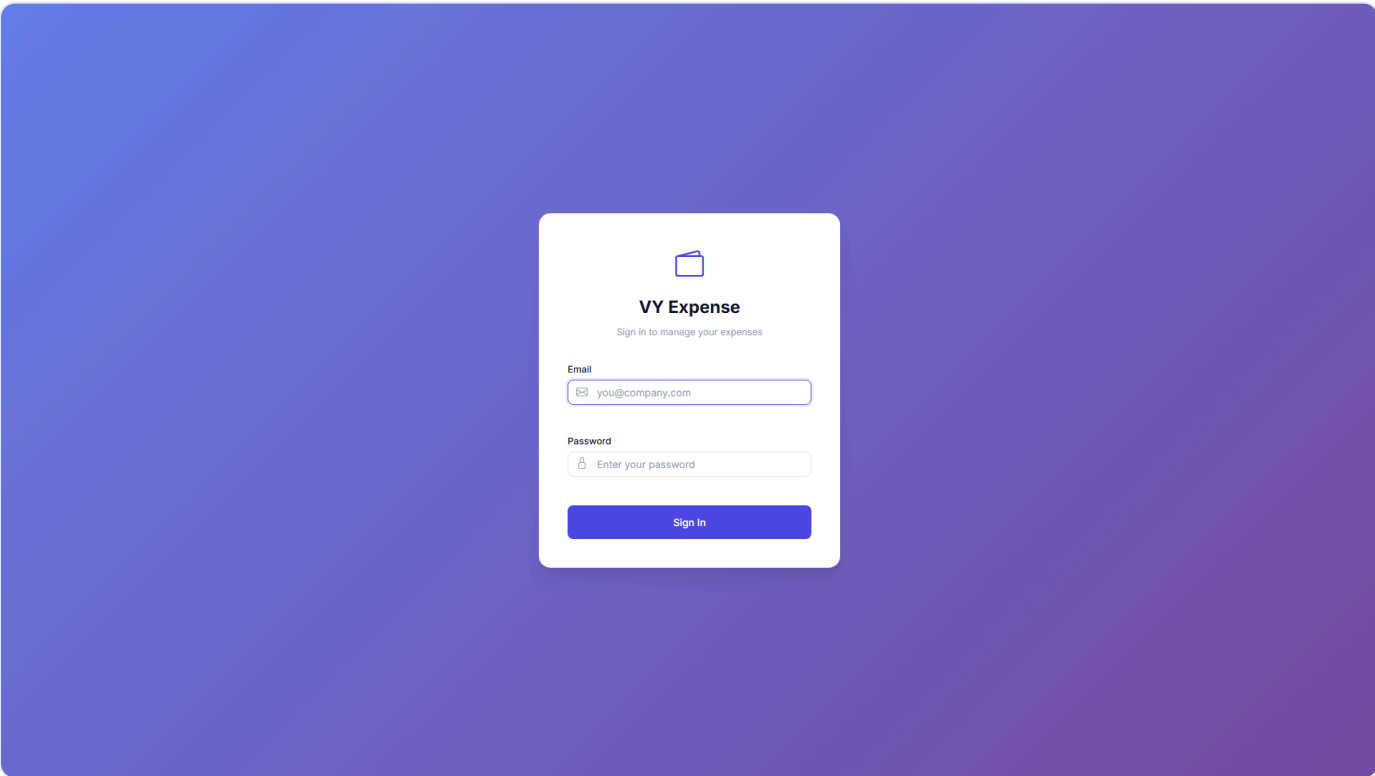
- <http://localhost:5000/app/login>

Optional sample users and data

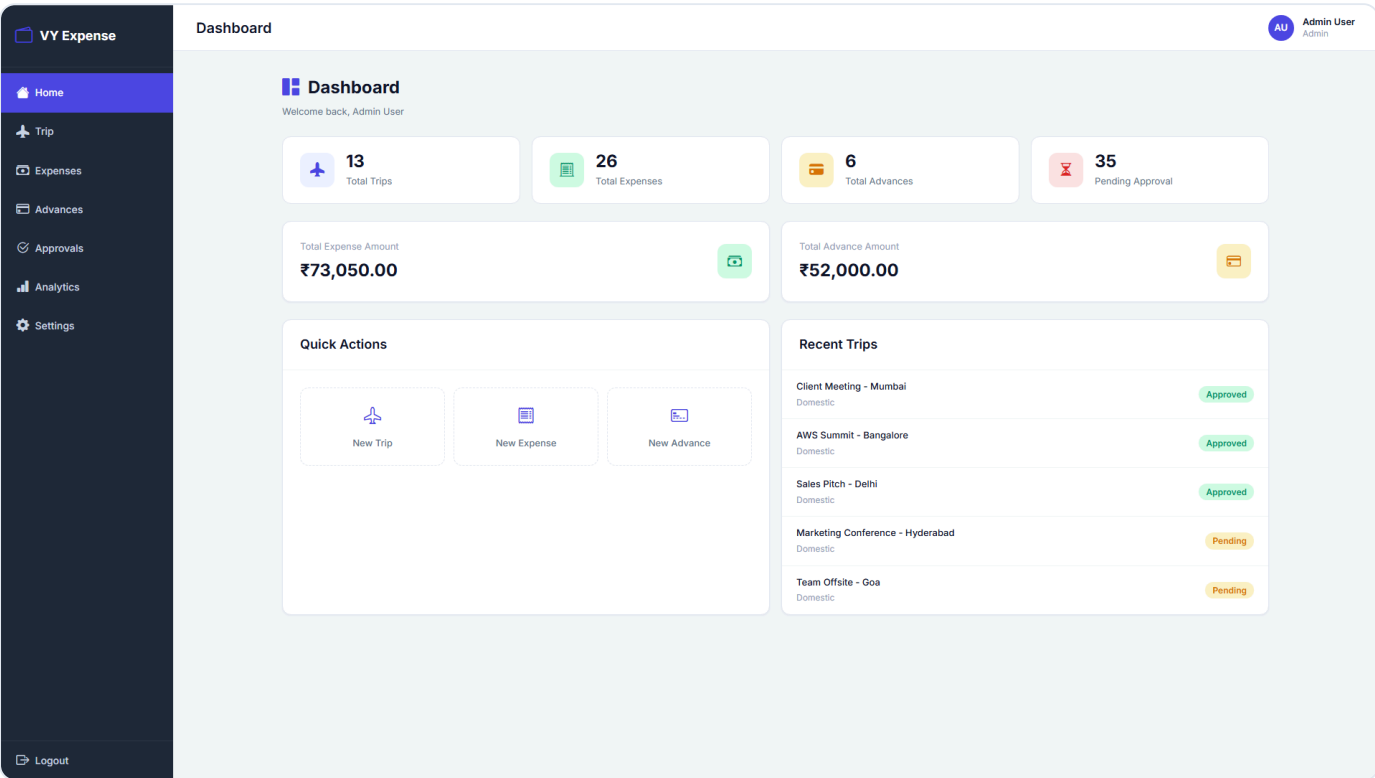
```
node database/create-users.js
node database/seed-corporate-data.js
```


13. Visual Project Walkthrough (Screenshots)

13.1 Login



13.2 Dashboard



13.3 Trips List

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

Trips

AUAdmin UserAdmin

+ New Trip

+ Trips

Manage your business trips

AllPendingApprovedRejectedFinished

SUBMITTER	TRIP NAME	TRAVEL TYPE	STATUS	CREATED	ACTIONS
Rajesh Kumar	Client Meeting - Mumbai	Domestic	Approved	17 Feb 2026	
Deepa Nair	AWS Summit - Bangalore	Domestic	Approved	17 Feb 2026	
Priya Sharma	Sales Pitch - Delhi	Domestic	Approved	17 Feb 2026	
Arun Patel	Marketing Conference - Hyderabad	Domestic	Pending	17 Feb 2026	
Rajesh Kumar	Team Offsite - Goa	Domestic	Pending	17 Feb 2026	
Priya Sharma	Vendor Negotiation - Singapore	International	Approved	17 Feb 2026	
Suresh Reddy	Tech Workshop - Pune	Domestic	Rejected	17 Feb 2026	
Meera Iyer	Office Inspection - Local	Local	Approved	17 Feb 2026	
Vikram Singh	Finance Audit - Chennai	Domestic	Approved	17 Feb 2026	
Arun Patel	Product Launch - London	International	Pending	17 Feb 2026	
Regular User	ewrwertwe4tw	Domestic	Pending	16 Feb 2026	
Regular User	ewrwertwe4tw	Domestic	Pending	16 Feb 2026	
Regular User	Test Trip	Domestic	Finished	16 Feb 2026	

13.4 Trip Details

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

TripDetails

AUAdmin UserAdmin

Client Meeting - Mumbai

domestic trip

Approved

TYPEDomestic

STATUSApproved

CREATED17 Feb 2026

TOTAL EXPENSE₹21,550

TOTAL ADVANCE₹15,000

Itinerary

Expenses (4)

	TITLE	AMOUNT	CATEGORY	STATUS	DATE	ACTIONS
☐	—	₹3,500	—	Pending	—	
☐	—	₹5,200	—	Pending	—	
☐	—	₹12,000	—	Pending	—	
☐	—	₹850	—	Pending	—	

Advances (1)

	AMOUNT	PAID THROUGH	REFERENCE	STATUS	ACTIONS
☐	₹15,000	Bank Transfer	—	Pending	

All expenses and advances must be approved/rejected before finishing.

Finish Trip

13.5 New Trip Form

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

NewTrip

AUAdmin UserAdmin

New Trip

Create a new trip request

Trip Name *

eg: Trip to New York

Trip Description

Describe the purpose of this trip

Select Users

Suresh Reddy (suresh.reddy@nexacorp.com)
Vikram Singh (vikram.singh@nexacorp.com)
Meera Iyer (meera.iyer@nexacorp.com)
Admin User (admin@nexacorp.com)

Hold Ctrl/Cmd to select multiple users.

Travel Type *

Local

You are creating this request for a trip
within your company locality.

Domestic

You are creating this request for a trip
within your home country.

International

You are creating this request for a trip
outside your home country.

TRIP ITINERARY

+Flight

+Bus

+Train

+Cab

Cancel

Save and Submit

13.6 Expenses List

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

Expenses

Expenses

Track and manage expenses

+ New Expense

AllPendingApprovedRejected

TITLE	AMOUNT	CATEGORY	TRIP	STATUS	DATE	ACTIONS
Vendor Lunch Meeting	₹120	Meals	Vendor Negotiation - Singapore	Pending	16 Mar 2026	
Marina Bay Sands - 4 nights	₹1,200	Accommodation	Vendor Negotiation - Singapore	Pending	15 Mar 2026	
Flight Chennai-Singapore	₹850	Travel	Vendor Negotiation - Singapore	Pending	15 Mar 2026	
Grab rides	₹80	Fuel	Vendor Negotiation - Singapore	Pending	15 Mar 2026	
Local Travel	₹400	Travel	Finance Audit - Chennai	Pending	25 Feb 2026	
Audit Documents Printing	₹950	Supplies	Finance Audit - Chennai	Pending	25 Feb 2026	
Working Meals - 3 days	₹1,500	Meals	Finance Audit - Chennai	Pending	25 Feb 2026	
Client Entertainment	₹8,000	Entertainment	Sales Pitch - Delhi	Pending	21 Feb 2026	
Airport Parking	₹600	Parking	Sales Pitch - Delhi	Pending	20 Feb 2026	
Flight Chennai-Delhi	₹6,800	Travel	Sales Pitch - Delhi	Pending	20 Feb 2026	
ITC Maurya - 2 nights	₹15,000	Accommodation	Sales Pitch - Delhi	Pending	20 Feb 2026	
Printed Proposals	₹1,200	Supplies	Sales Pitch - Delhi	Pending	20 Feb 2026	
Cab to branch office	₹350	Fuel	Office Inspection - Local	Pending	18 Feb 2026	
Stationery Restock	₹750	Supplies	Office Inspection - Local	Pending	18 Feb 2026	
Working Lunch	₹300	Meals	Office Inspection - Local	Pending	18 Feb 2026	
Team Dinner Day 2	₹1,800	Meals	AWS Summit - Bangalore	Pending	17 Feb 2026	
Working Lunch Day 1	₹450	Meals	AWS Summit - Bangalore	Pending	16 Feb 2026	
Test Meal	₹250	Travel	Test Trip	Approved	16 Feb 2026	
Train Chennai-Bangalore	₹1,200	Travel	AWS Summit - Bangalore	Pending	15 Feb 2026	
Conference WiFi Pass	₹500	Communication	AWS Summit - Bangalore	Pending	15 Feb 2026	
Lemon Tree Hotel - 3 nights	₹9,000	Accommodation	AWS Summit - Bangalore	Pending	15 Feb 2026	
Terst	₹200	Communication	Test Trip	Approved	15 Feb 2026	
Client Dinner at Wasabi	₹3,500	Meals	Client Meeting - Mumbai	Pending	11 Feb 2026	
Hotel Taj Lands End - 2 nights	₹12,000	Accommodation	Client Meeting - Mumbai	Pending	10 Feb 2026	
Flight Chennai-Mumbai	₹5,200	Travel	Client Meeting - Mumbai	Pending	10 Feb 2026	
Airport Cab	₹850	Fuel	Client Meeting - Mumbai	Pending	10 Feb 2026	

13.7 New Expense Form

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

NewExpense

AUAdmin UserAdmin

New Expense

Record a new expense

Title *

e.g. Hotel stay — Mumbai

Amount *

₹ 0.00

Currency

INR — Indian Rupee

Date *

03/30/2026

Category *

Select category

Trip *

Select trip

Description

Optional details about this expense

Cancel

Create Expense

13.8 Advances List

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

Advances

AUAdmin UserAdmin

+ New Advance

Manage cash advance requests

AllPendingApprovedRejected

AMOUNT	CURRENCY	PAID THROUGH	REFERENCE	TRIP	STATUS	ACTIONS
₹15,000	INR	Bank Transfer	—	Client Meeting - Mumbai	Pending	
₹8,000	INR	Bank Transfer	—	AWS Summit - Bangalore	Pending	
₹20,000	INR	Company Card	—	Sales Pitch - Delhi	Pending	
₹2,000	USD	Wire Transfer	—	Vendor Negotiation - Singapore	Pending	
₹2,000	INR	Cash	—	Office Inspection - Local	Pending	
₹5,000	INR	Bank Transfer	—	Finance Audit - Chennai	Pending	

13.9 New Advance Form

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

NewAdvance

AUAdmin UserAdmin

New Advance

Request a cash advance

Amount *

₹ 0.00

Currency

INR — Indian Rupee

Paid Through *

Select method

Reference Number

Transaction / cheque ref

Trip *

Select trip

Notes

Additional details about this advance

Cancel

Request Advance

13.10 Approvals Module

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

Approvals

AUAdmin UserAdmin

Approvals

Review and approve submissions

Trips

Expenses

Advances

All

Pending

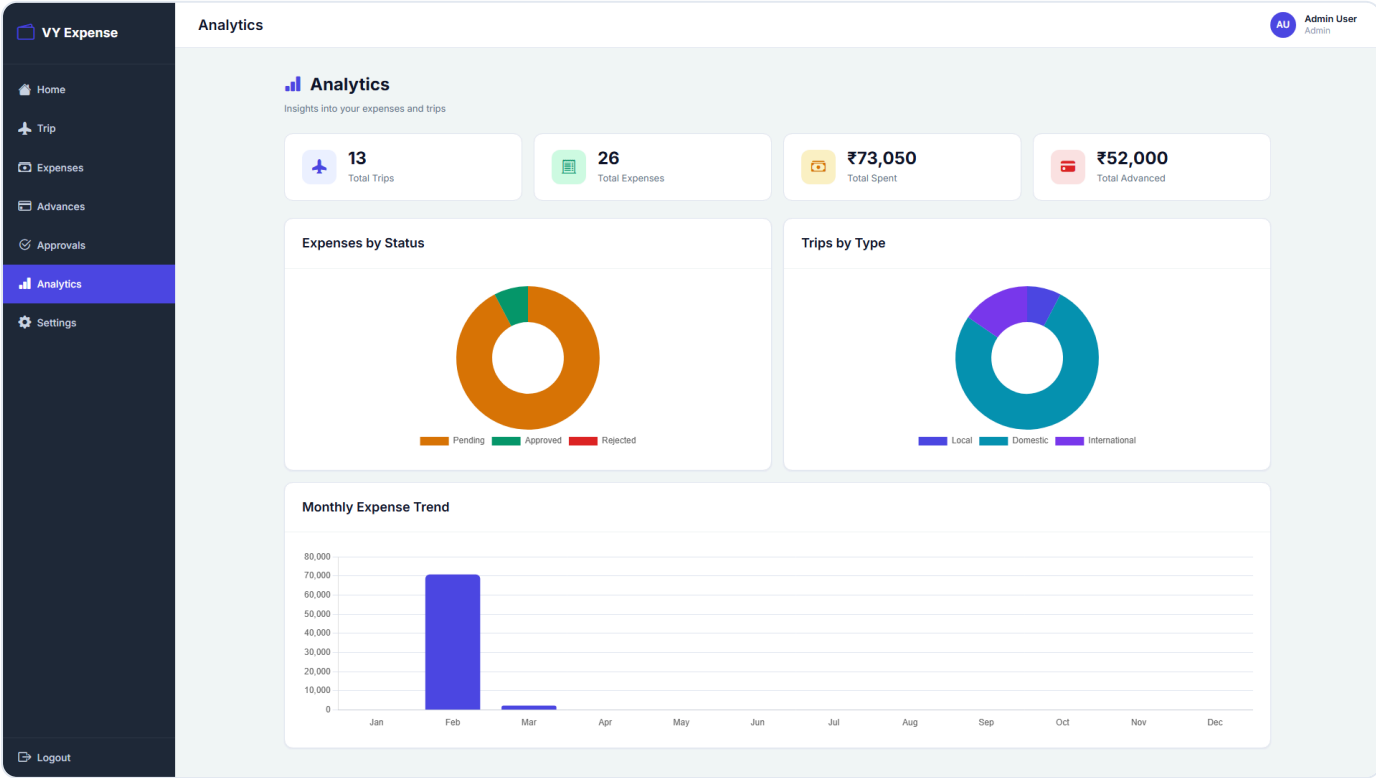
Approved

Rejected

Finished

☐	TRIP NAME	TYPE	STATUS	CREATED
☐	Marketing Conference - Hyderabad	Domestic	Pending	17 Feb 2026
☐	Team Offsite - Goa	Domestic	Pending	17 Feb 2026
☐	Product Launch - London	International	Pending	17 Feb 2026
☐	ewrwertwe4tw	Domestic	Pending	16 Feb 2026
☐	ewrwertwe4tw	Domestic	Pending	16 Feb 2026

13.11 Analytics Module



13.12 Settings and User Management

VY Expense

Home

Trip

Expenses

Advances

Approvals

Analytics

Settings

Logout

SettingsAdmin UserAdmin

SettingsManage your profile and preferences

Profile

NameAdmin User

Emailadmin@test.comEmail cannot be changed

Roleadmin

Save Changes

Change Password

Current Password

New Password

Confirm New Password

Update Password

Expense Categories

AccommodationCommunicationEntertainmentFuelMealsOtherParkingSuppliesTravel

New category nameAdd

Manage Users

Suresh Reddy suresh.reddy@nexacorp.com	SUBMITTER
Vikram Singh vikram.singh@nexacorp.com	APPROVER
Meera Iyer meera.iyer@nexacorp.com	ADMIN
Admin User admin@nexacorp.com	ADMIN
Rajesh Kumar rajesh.kumar@nexacorp.com	SUBMITTER
Priya Sharma priya.sharma@nexacorp.com	SUBMITTER
Arun Patel arun.patel@nexacorp.com	SUBMITTER
Deepa Nair deepa.nair@nexacorp.com	SUBMITTER
Admin User admin@test.com	ADMIN
Regular User user@test.com	SUBMITTER

Add User

Name

Email

Password

User

Add User

14. API Endpoint Summary

Area	Method	Endpoint	Purpose
Auth	POST	/api/auth/login	Login and session creation
Auth	POST	/api/auth/logout	Destroy session
Auth	GET	/api/auth/me	Current session user
Trips	GET	/api/trips	List trips by role/filter
Trips	POST	/api/trips	Create trip with itinerary
Trips	POST	/api/trips/:id/status	Approve/Reject trip
Trips	POST	/api/trips/:id/finish	Mark trip finished
Expenses	GET	/api/expenses	List expenses
Expenses	POST	/api/expenses	Create expense
Expenses	POST	/api/expenses/approve	Bulk approve
Expenses	POST	/api/expenses/:id/reject	Reject with reason
Advances	GET	/api/advances	List advances
Advances	POST	/api/advances	Create advance
Advances	POST	/api/advances/approve	Bulk approve
Advances	POST	/api/advances/:id/reject	Reject with reason
Analytics	GET	/api/analytics/data	Chart/aggregation payload

15. Testing and Quality

Current project test assets include:

- backend model/controller/route tests under tests/
- frontend tests under frontend/tests/

Recommended regression checks:

- login/logout session behavior
- role guard behavior (submitter vs admin)
- trip status transitions
- approved-trip-only creation for expenses/advances
- analytics data validity after approvals

16. Suggested Enhancements

1. Add JWT + refresh token option (if stateless scaling is needed).
2. Add audit history table for all status transitions.
3. Add object storage option for receipts (S3/Azure Blob) for large files.
4. Add E2E tests for major user journeys.
5. Add OpenAPI spec generation for API contracts.
6. Add structured logging + request IDs.

17. Conclusion

The application is a complete corporate expense-travel workflow platform with:

- robust relational data model
- clear approval lifecycle
- full frontend experience for submitters and admins
- practical analytics for operational insight

This documentation package includes architecture and ER vector diagrams, live walkthrough screenshots, and complete technical exploration of how the codebase works end to end.